

WIMBORNE AGCM, AB, Canada

WMO: 719040

Lat: 51.9358N

Lon: 113.5908W

Elev: 940

StdP: 90.53

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-29.0	-26.1	-32.4	0.2	-28.7	-29.5	0.3	-25.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	27.8	16.3	26.1	15.7	24.4	15.2	18.7	24.6	17.5	23.6	16.4	22.4	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.4	13.1	22.1	15.1	12.0	20.7	13.9	11.1	19.3	57.0	24.4	53.0	23.7	49.5	22.4	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	DB	-34.1	31.5	2.3	2.0	-35.7	32.9	-37.1	34.1	-38.3	35.2	-40.0	36.6	(4)
(5)			WB	-34.2	21.3	2.2	1.6	-35.8	22.4	-37.1	23.4	-38.4	24.3	-40.0	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-8.0	-9.7	-3.8	3.1	9.9	13.7	16.2	15.5	11.0	4.1	-3.7	-9.7	(6)	
	DBStd	11.20	8.14	8.09	7.65	5.10	4.42	2.95	2.49	2.95	4.58	5.09	7.21	7.77	(7)	
	HDD10.0	3066	557	552	427	211	57	6	0	1	41	193	411	611	(8)	
	HDD18.3	5508	815	786	686	457	262	141	74	95	221	442	660	869	(9)	
	CDD10.0	619	0	0	0	4	54	116	192	172	71	9	1	0	(10)	
	CDD18.3	19	0	0	0	0	1	2	8	7	2	0	0	0	(11)	
	CDH23.3	597	0	0	0	3	51	62	181	198	99	3	0	0	(12)	
	CDH26.7	108	0	0	0	0	4	6	33	43	21	0	0	0	(13)	
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)	
(15) Precipitation	PrecAvg	516	19	16	25	38	68	99	73	63	45	26	24	20	(15)	
	PrecMax	678	37	32	54	87	110	254	160	114	110	51	60	44	(16)	
	PrecMin	327	4	3	8	15	24	44	28	12	7	8	5	6	(17)	
	PrecStd	68	8	7	10	17	25	39	31	24	11	12	12	10	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	7.9	14.2	22.1	27.2	27.2	29.8	30.4	29.3	22.7	16.3	8.0	(19)	
		MCWB	3.0	3.5	6.3	9.9	12.9	15.8	18.2	16.7	14.5	10.8	8.3	3.1	(20)	
	2%	DB	6.5	5.2	10.5	18.5	24.8	25.2	27.3	27.5	26.2	18.2	11.4	4.9	(21)	
		MCWB	2.4	1.6	4.5	8.5	11.9	15.2	18.1	16.7	13.8	9.1	5.4	1.7	(22)	
	5%	DB	5.0	3.1	7.8	15.0	22.1	23.1	25.3	25.5	23.6	15.4	7.9	3.2	(23)	
		MCWB	1.6	0.2	3.1	6.4	10.9	14.0	17.4	16.3	13.2	8.0	3.3	0.1	(24)	
	10%	DB	3.3	1.3	5.4	11.9	19.5	21.2	23.5	23.5	20.4	12.5	5.4	1.0	(25)	
		MCWB	0.4	-1.0	1.7	4.8	9.9	12.9	16.5	15.7	11.8	6.4	1.8	-1.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.5	7.4	10.4	13.7	18.2	21.0	19.6	15.8	11.9	8.2	3.3	(27)	
		MCDB	7.2	7.3	13.3	21.4	24.8	25.0	26.6	25.4	25.6	21.1	15.3	6.5	(28)	
	2%	WB	2.6	1.8	5.0	8.6	12.5	16.3	19.3	18.3	14.6	10.0	5.6	1.7	(29)	
		MCDB	5.9	4.8	9.8	17.9	22.5	22.9	25.0	24.6	24.1	17.0	10.5	4.8	(30)	
	5%	WB	1.7	0.4	3.3	6.9	11.5	15.0	18.0	17.2	13.5	8.4	3.8	0.2	(31)	
		MCDB	4.9	2.8	7.5	13.8	20.5	20.9	24.1	23.8	21.8	14.5	7.5	2.9	(32)	
	10%	WB	0.3	-0.8	1.9	5.3	10.5	13.8	16.9	16.0	12.3	7.0	2.1	-1.4	(33)	
		MCDB	3.2	1.1	5.1	11.0	18.4	19.9	22.5	22.4	19.9	11.9	4.7	1.1	(34)	
(35) Mean Daily Temperature Range	MDBR		9.7	10.1	10.3	11.8	13.3	12.2	12.7	13.5	13.7	12.1	10.1	9.5	(35)	
	5% DB	MCDBR	9.9	11.6	13.0	18.3	18.3	16.2	16.5	18.1	20.1	16.7	13.4	11.2	(36)	
		MCWBR	7.0	8.6	8.1	9.7	8.1	7.8	9.1	9.2	9.4	9.2	8.5	8.4	(37)	
	5% WB	MCDBR	10.0	11.2	12.6	17.2	16.0	14.1	15.6	16.1	18.3	16.0	13.0	11.0	(38)	
		MCWBR	7.4	8.5	8.3	9.3	7.5	7.5	9.1	8.9	8.7	8.9	8.7	8.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.259	0.270	0.284	0.325	0.337	0.341	0.342	0.356	0.309	0.280	0.251	0.241	(40)	
	taud		2.342	2.404	2.449	2.366	2.389	2.408	2.419	2.372	2.471	2.536	2.480	2.349	(41)	
	Ebn at Noon		766	869	923	914	915	911	904	871	883	851	787	730	(42)	
	Edh at Noon		60	75	88	109	112	111	108	107	86	66	51	49	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.09	2.12	3.50	4.72	5.71	6.08	6.45	5.25	3.67	2.21	1.16	0.83	(44)	
	RadStd		0.09	0.17	0.28	0.40	0.38	0.38	0.33	0.37	0.39	0.26	0.10	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page